Lichen

L2s In CHannels ENforced by bitcoin

waxwing

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Abstract

A *contract* with a hub lets a user hold the hub's tokens on a layer 2 and withdraw them as bitcoin without trusting the hub for honesty or liveness. The L2 tokens are collateralised by the bitcoin in the user-hub multisig output[1], and if the hub refuses a withdrawal, or cannot answer because it is offline, it is enforced on Bitcoin. The only assumptions are those of the layer 2 itself, kept apart from the hub. No change to Bitcoin's consensus is required. A withdrawal is a claim that a program, the withdrawal statement, ran on some input and produced an output: the statement is proved with RISC Zero and wrapped in Groth16, and a disputed claim is settled on Bitcoin by a verification gadget that checks the Groth16 proof: BABE, by witness encryption and garbled circuits, in four or five transactions; or BitVMX, the first one built, by an interactive search. Withdrawals are executed cooperatively in the contract — with overwhelming likeliness because the statement is deterministically verifiable in milliseconds — or, otherwise, a refused one is played as a dispute on chain, set up at the start of each contract. The trustlessness gain is paid for with liquidity (with receiving ability only contingent on offered "inbound" from the hub, as in Lightning). The hub (but not user) must also stay live during normal operation, since an offline hub loses disputes by timeout. The design, including a privacy-focused shielded note L2, is built and measured in [9].

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1. Introduction

1.1. Obvious objections to Lichen and why they're mostly wrong

One central hub is dangerous. If the hub disappears, the user gets the money. The hub cannot Sybil attack users by printing tokens fictitiously; the honest users' money is all backed. The only remaining risks are inherent to the L2 itself (is it a validity rollup or a quorum? Does it inherit finality from Bitcoin or is finality weaker? Etc.)

Bridges usually mean you can only withdraw a fixed size utxo, meaning operators and relying on swaps. Surely that applies here? No. The withdrawal amount is encoded in bits in the bitcoin transaction witness, in dispute. This allows us to withdraw arbitrary amounts via multiple outputs of certain denominations [5.2].

Don't these constructions usually require a 1 of n honesty assumption from a signing committee? Not here: we avoid this by the LN-GAP [1] trick: the contract is a *bilateral* contract between the hub and each user; the presigning happens between them, just as in Lightning.

But if a bunch of users open contract with a hub, they can just .. pay each other through Lightning channels? A correct critique, partly, if the L2 ledger doesn't do anything but plain payments. Even then, you could avoid receiver liveness being needed, avoid routing and other minor things. But the real answer is: the L2 can offer more functionality. It can use zcash style shielded notes, allowing very high privacy while staying in the L2. It can also (in a different "lichen" on a different hub) offer rich smart contracting, etc.

But this is an IOU/ hub's token, so it's not actually bitcoin. Incorrect with a crucial caveat. See the first point: every sat is redeemable via the user-hub contracts, *without* the hub's permission, one-to-one, so these are actually bitcoin *but only as long as you have liquidity to receive them.* A new user on L2 without a contract with the hub, or a user holding a balance

on L2 far in excess of their inbound liquidity in their own contract, is holding *credit*, and has no such guarantee. But that's their choice.

But still, a single central hub feels like bad design. Why not multiple hubs? Having multiple hubs *will* make sense — see Section 8, but done naively, it is broken. It's because of what these L2 tokens are. They are fungible, enforceable claims on the hub's money; they are minted by the hub itself, just on its own say-so. If you had hub 1 and hub 2, those claims are not transferrable trustlessly to the hubs themselves. If hub 2 prints a billion sats out of nothing and users accept it on the L2 they can redeem it with hub1, leaving hub1 out of pocket. From this line of thinking we see an interesting comparison with ecash servers: the hub can print as much as it likes, but it cannot steal/"rug".

So what's the catch? The negatives are twofold: one is liquidity, and this is easy to understand as the analog of the inbound liquidity pain point in Lightning, because it's basically the same thing. Such a hub *can* choose to give no inbound liquidity to a user: they get tokens on L2 to spend, but they cannot (safely) receive. If they receive, it's just credit that someone else will hopefully buy from them. But the design does not assume that, of course, it allows receiving up to a "cap": the user negotiates that extra potential receiving amount (inbound liquidity) when it sets up the contract with the hub. The second is the on-chain dispute, whose size depends on the gadget: four or five transactions and 10–31 kvB under BABE, lichen's gadget, against tens of transactions and 100–200 kvB under BitVMX, the first, both as measured on Mutinynet [4, 3].

1.2. A note on the delta to LN-GAP

LN-GAP [1] puts contracts governed by arbitrary programs into Lightning [13] channels: a contract's outcome is decided by a game whose moves are signed off-chain, with pre-signed transactions that let either party take a dispute to Bitcoin [1]. Lichen mainly differs in trying to achieve trustlessness at the expense of a potentially larger (though still O(1) in the size of the withdrawal statement) onchain resolution; a tradeoff that fits well with the intent of creating *an L2 from a bunch of contracts with a hub*:

	LN-GAP	Lichen w/ BABE
venue	trusted: a majority of members attests moves	none
a dispute is decided by	the venue's attestations	the chain: a challenge, answered from garbled circuits
suited to	interactive games where time has value	withdrawals from a hub's layer 2
on-chain cost of a dispute	constant	constant: about 30 kvB at most (BitVMX: log of the verifier's run)

Why Lichen needs no venue. It's more about choice than logical requirement. In LN-GAP the parties *could* just play ForceMove and accept that the game goes onchain when someone is not cooperative. That's not much fun for a 100 move chess game or similar. However the tradeoff of trust vs length of game is drastically different for Lichen: the game length is fixed in advance, and with BABE it is short: four or five transactions. While for such an L2 to be

viable it's absolutely crucial to try to remove third party trust. That's why the "venue" concept is removed here.

2. Contracts with the hub

Alice, the user, has a contract with a **Hub**, instantiated as a 2 of 2 output (see Section 5 for the possibility of embedding this inside a Lightning channel), with these terms:

D	Alice's deposit: she puts D bitcoin in and receives D of the hub's tokens on the L2
$V_{\max}$	the most she can withdraw: her D and the hub's liquidity $V_{\max} - D$
r_A, r_H	the *contract reserves*, Alice's and the hub's: returned in every honest outcome, forfeited to the other side by losing a dispute (Section 5.2); the contract holds $V_{\max} + r_A + r_H$
b	the amount she withdraws, by returning b tokens to the hub on the L2 with the memo c
c	the contract's identifier, its memo
T_{close}	the contract's end: with no withdrawal by then, the hub takes it

Every token is a liability of the hub, collateralised by the contract of whoever holds it, within that holder's cap; within $V_{\max}$ and before T_{close}, Alice trusts the hub for nothing, not even to be online: a withdrawal it does not answer is paid by timeout. The hub's provided liquidity is the sum of its users' $V_{\max} - D + r_H$.

Opening is a swap. Alice's D goes into the contract and her D tokens onto the L2, and neither may come without the other: a hub that took the funding and never minted would leave her nothing to return, and the default at T_{close} would hand it her D; a hub that minted first could be left with tokens backed by no contract. Alice first pays a deposit output, the hub's key and hers together, or hers alone after a timeout. The hub then mints D to the key $P = H(A \,\|\, T) \cdot A + T$, where A is Alice's L2 key and $T = t \cdot G$ a point of the funding transaction's own (the coefficient keeps Alice from choosing A against T), and Alice, once she sees that mint sealed, signs the funding's deposit input as an adaptor pre-signature locked to T. The hub can spend the deposit only by completing it, which publishes t and so unlocks her tokens. The hub's side needs no refund: if it never broadcasts, Alice refunds the deposit, and the locked tokens, which cost nothing to print, stay where they are. Its one risk is timing: t is public once the funding is in the mempool, so it broadcasts only well before the deposit's refund matures.

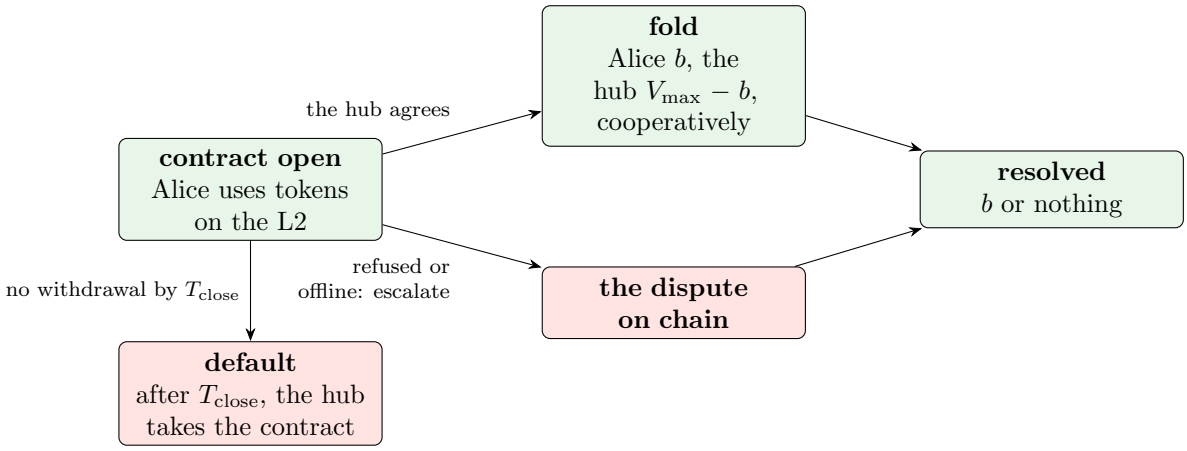


Figure 1: The life of a contract. Green states stay off-chain or settle cooperatively; red states are on chain.

3. How a withdrawal from the L2 is proved valid

3.1. The statement proved

For brevity we have not yet given a description of what such an L2 must look like, but its main features are here, to explain the proof structure.

The L2 is a ledger of notes, with state updates signed by a sequencer[2]. A note is an owner, a value, a memo and a random ρ, which makes every note distinct (two equal payments to the same owner are still two notes); its commitment is a SHA-256 hash of these, and the commitments form an append-only Merkle tree. A spent note is not removed from the tree, which would change every other note's path, but recorded by its *nullifier*, a hash of its commitment, in a set the sequencer keeps: spending a note twice repeats its nullifier. So a note's path under a root stays valid however the ledger moves on. In case we want *shielded* transactions, as implemented in [9], ρ also hides the note's contents, and the nullifier is derived from a secret key so that nobody can link it to its note. A state root SR hashes the height, the tree's root and the nullifier set's root, and the L2's sequencer signs it (BIP340).

With this context, we can understand what the withdrawal proof actually does. The user provides:

1. A note owned by the hub, of value b and memo c, marked as "withdrawal" is in the commitment tree of a state root SR. This is a Merkle path from the note to the root.

2. The sequencer published that root. SR carries the sequencer's signature, for its L2 height.

3. Validity back to genesis. A chain receipt (the sequencer's recursive proof) shows that every L2 block from genesis to now was a valid state transition. The same receipt covers anchoring: each anchor spends the previous anchor's output, starting from the L2's genesis seal. Each is in a real Bitcoin block, by Merkle path to that block's header. The last[3]

[2]We'll talk more about achieving finality via Bitcoin or otherwise later [6].

[3]See Section 6; this needn't be the last SR in time, but only a unique history, because (currently) we are using

anchor committed exactly SR, at its height. So SR is the root of the one anchored history.

4. Finality. Bitcoin headers follow on from the anchor's block, each meeting its own difficulty target, with total work of at least W. W is fixed when the contract opens, about 10 blocks' worth.

The result is the amount b, the memo c and $\mathcal{D}$, which identifies the contract: the sequencer's and hub's keys, the genesis output and W. So a proof for another deployment, history or finality threshold is a proof of a different statement. Under BitVMX the public output is $b \,\|\, c \,\|\, \mathcal{D}$: Bitcoin reads b for the payout, and checks c and $\mathcal{D}$ against the contract's terms. Under BABE, whose public input is fixed when the contract opens, the statement also checks the hub's one-time signature on b, and its public output is $c \,\|\, \mathcal{D} \,\|\, H(pk_H)$, the same for every amount (Section 4); with the exit (Section 7) it is $c \,\|\, \mathcal{D} \,\|\, H(pk_H \,\|\, O_D)$, O_D the contract's deposit coin. Since the chain proof's program id is built into the withdrawal program, the one program id the contract pins covers both.

Not fully addressed yet, however:

- This is not (yet) a proof of heaviest chain; that would require a counter-proof mechanism. It's currently judged that building a 10 block fork is expensive enough to make attack uneconomic.
- Sequencer going quiet, or censoring user transfers: the exit (Section 7), in which Alice proves owning notes at an anchored state and the hub can counter-prove a later spend, is built and played on regtest, not yet on real proofs.
- The data availability question has not been addressed; the exit is safe exactly when the user can read the anchored data she needs.

3.2. From a program to a verifiable claim

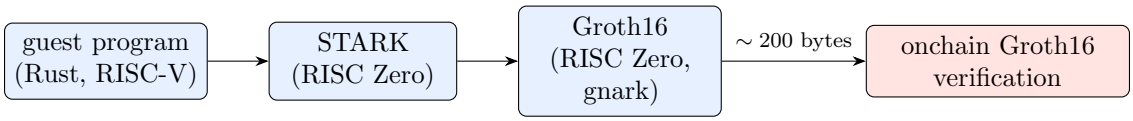


Figure 2: The withdrawal statement runs in RISC Zero's zkVM; its proof is compressed to a Groth16 proof, which the verification gadget checks on Bitcoin (Section 4) [17]. An explainer covers each stage, under both gadgets [2].

What the contract must bind. Whatever gadget settles a dispute, the contract must tie the disputed proof to three things: the program, since a valid proof of another program proves nothing; this contract's terms, its memo c and identifier $\mathcal{D}$ (the deployment's keys, the genesis seal and the work W), so that a proof for another contract, deployment, history or finality threshold fails; and the amount b, in a form Script can read, for the payout (Section 5.2). With RISC Zero, the proof's public input is a digest of the program's image id and its journal $b \,\|\, c \,\|\, \mathcal{D}$; because RISC Zero's Groth16 key is the same for every program, the image id must be checked explicitly (one suffices, since the withdrawal program has the chain program's id built in). How

a special withdrawal transaction, not any arbitrary note

the binding is enforced is the gadget's business. Under BitVMX, Alice signs the verifier's input word by word and the contract checks the image id, c and $\mathcal{D}$, and reads b, from those signatures [3, §2]. Under BABE, the image id, c and $\mathcal{D}$ are fixed when the contract is set up, and b, which cannot be, is bound by the hub's signature on the amount Alice asserts [4, §5].

4. Settling a dispute on Bitcoin

The contract is a single taproot output, 2-of-2 between Alice and the hub, whose leaves hold the ways it can end; transactions pre-signed at opening let either party carry out its side of a dispute without the other. What settles a dispute is an on-chain *verification gadget*.

The idea is, given a proof whose public input binds the program's context ($b \,\|\, c \,\|\, \mathcal{D}$ as mentioned above), adjudicate the claim on Bitcoin. The context is checked, the amount b is read into Script, and the loser pays. To emphasize: this is all verification. The proving side, up to the Groth16 proof, is the same whichever gadget is used, but for the statement's public output and BABE's check of the hub's signature on b.

4.1. Values that Script can read: one-time signatures

Bitcoin Script can check a signature only on the spending transaction, never on a message of the signer's choosing. A contract that must act on a value a party chooses later, above all Alice's b, uses Winternitz one-time signatures, which Script verifies by hashing alone, leaving the signed digits on the stack to compare [14, 10]. The keys are exchanged when the contract opens, one per value, each used once; signing two different values under one key is *equivocation*, which a leaf can prove, giving the signer's stake to its counterparty. Every gadget below reads b this way.

4.2. The first gadget: BitVMX

BitVMX [10] settles a claim about a RISC-V program's run by an interactive search over its execution. Alice claims that the Groth16 verifier, run on her signed input, halts with success; if the hub disagrees, they narrow the run down, eight segments a round, to a single step, which Alice then proves in Script or the hub disproves. Each move is a pre-signed transaction carrying the mover's one-time signatures, and a lie about a memory read is caught through an authenticated jump list instead of a second search. It was built first, has been played to the end on Mutinynet, and is still built; the companion paper [3] gives the transaction graph and the measurements.

4.3. Lichen's gadget: BABE with authenticated amounts

BABE [5] replaces the search with a challenge. At setup, the hub garbles a small circuit that turns the first element of Alice's proof into the key of a witness encryption, so that only a valid proof unlocks the message she needs to be paid. Alice posts that element, signed; if the hub challenges, it must reveal the circuit's input labels for it, and with a valid proof Alice decrypts and posts the message. Against a hub that garbles wrongly, Alice checks the garbling by cut-and-choose: the hub makes 78 instances, she opens 68 and keeps 10, and lichen replaces BABE's soldering with verifiable secret sharing, so that the setup takes about 19 seconds and the kept instances about 240 MiB. BABE's obstacle is that the statement is fixed at setup, while Alice chooses b only when she withdraws. The amount therefore moves into the witness, together with

the hub's one-time signature on it, which the hub must give on chain for exactly the amount Alice asserted, or lose by timeout. This is our proposal [7]. Unlike GOAT's cryptographic alternative [6], in our proposal, BABE's cryptography is unchanged. It is built, it is lichen's default gadget, and three of its endings have been played on Mutinynet with a real RISC Zero proof; a companion paper [4] gives the construction as built, a sketch of its security, and the measurements.

	BitVMX	BABE
exit against a vanished hub	23 kvB, 5 transactions	∼1.2 kvB, 2 (est.; played on regtest)
a true claim refused, played to the end	∼120 kvB, 23 transactions	9.7 kvB, 4; 30.5 kvB, 5 if the hub challenges
a false claim	100–200 kvB	29.9 kvB, 5
the worst case	∼200 kvB (a lie about a read)	30.5 kvB
pre-signed per contract	141 transactions	19 (20 with the exit)
setup per contract, beyond pre-signing	none	19 s on a laptop; ∼240 MiB stored by the user
contracts a hub can hold, disputes lasting a week*	3,300 (8,350 with a window per move)	∼47,000

Table 1: The two gadgets, as measured on Mutinynet but where marked [3, 4]. *Before congesting the chain becomes a profitable attack: for N contracts, every move due within a window of w blocks must fit in it, $Ns \leq w \cdot 1$ MvB [19], and the dispute lasts the sum of its windows. Measured before BABE's graph carried the exit, which adds about 5 kvB to a refused claim and brings the contracts a hub can hold in a week to about 40,000 (est.; Section 7).

5. The contract logic

5.1. Three ways out

A contract ends in one of three ways; its output carries three leaves (not including a revocation if inside a channel):

- the *fold*: a cooperative update of the channel (Alice $b + r_A$, the hub $V_{\max} - b + r_H$), the normal case, nothing on chain;
- `default`: after T_{close}, a transaction pre-signed at the contract's opening pays the hub $V_{\max} + r_H$ and returns Alice's reserve r_A. This is the liveness requirement on Alice: turn up at the end, i.e. it's a very low bar requirement; whereas the hub must always be ready to answer disputes.
- `request` (BitVMX's `escalate`): Alice's claim goes on chain as the first step of the dispute. This will almost never happen, because given Alice's proof, the hub can calculate ∼ immediately that he's guaranteed to lose (since Groth16 verification is fast), so just pays out cooperatively (*fold* as above).

On its own, or in a channel? A contract is a standalone output, a 2-of-2 between Alice and the hub, opened by one on-chain transaction and folded by another. This is the design as built, and the one this paper assumes. The same output could sit in a Lightning channel commitment, making the fold a channel update and letting deposits arrive as Lightning payments, as earlier LN-GAP code demonstrates. But the price is high. Every channel update re-signs the contract's pre-signed transactions for both commitments (20 each under BABE, 141 under BitVMX); renewal, though off chain, is still a new contract and, under BABE, a new setup; and a channel's revoked states bring back the always-on watchtower that a standalone contract avoids (Section 5.3). We treat it as a possible later optimisation, not part of the design.

5.2. Outcomes

The dispute only ever decides Alice's withdrawal: b if her claim stands, nothing if it fails; the rest of $V_{\max}$ goes to the hub. The reserves follow the dispute: the loser's goes to the winner, whether it lost on the merits, by signing two values of b (`equiv_b`), or by not answering in time, less the fee floors F of the dispute's pre-signed transactions, which the loser's reserve pays. So losing costs the two sides differently. Alice, ruled against, loses the whole contract, her reserve included. The hub, ruled against, loses its reserve and keeps $V_{\max} - b$, since b was all the dispute decided. Of the contract's $V_{\max} + r_A + r_H$:

outcome	Alice	the hub
fold	$b + r_A$	$V_{\max} - b + r_H$
`default`, no withdrawal by T_{close}	r_A	$V_{\max} + r_H$
Alice wins a dispute	$b + r_A + r_H - F$	$V_{\max} - b$
Alice loses a dispute	0	$V_{\max} + r_A + r_H - F$

Paying b: the bit outputs. When Alice wins, the payout must send her b, a value she chose when she claimed; but the payout is pre-signed, so its amounts are fixed, and Script cannot compute an amount. One pre-signed transaction per possible b would be too many. Instead, as for numeric DLCs [23], the payout decomposes b in binary. Amounts are counts of one satoshi, and a contract of K bits has the cap $V_{\max} = 2^K - 1$ sats; the payout has one output of 2^i sats for each bit i from 9 to $K - 1$, each with three leaves:

leaf	spent by	condition
Alice's	Alice, after a delay δ	her one-time signature on b has bit i set
the hub's	the hub	the same signature has bit i clear
`equiv_b`	the hub	two signatures on different values of b

Since b is under a one-time key, the bits Alice can claim are those of the single value she signed, and their outputs sum to b; the hub takes the rest. Bits below 9 have no output, so a withdrawal settled by dispute is paid in multiples of 512 sats, the remainder going to the hub. For $b = 5{,}120$ sats she takes the outputs for bits 10 and 12. Her leaves wait δ so that the hub can show an equivocation first. A contract of $K = 24$ bits has a cap of about 0.168 BTC and 15 bit outputs; each output's spend is about 330 vB, and each party can sweep its outputs in one transaction. The reserves sit in one more output, paid to the winner, with an `equiv_b` leaf

of its own. In the cooperative fold none of this appears: the fold simply pays b.

The contract reserve. As in Lightning, a penalty deters only a party with something to lose, and here that is not what each party put in but what each loses by losing a dispute. Without reserves, Alice risks only her honest outcome, the return she could actually prove, which is nothing once she has spent her tokens: a false claim of $V_{\max}$ is then a free option on the hub being offline. The hub makes no claims, only refusals, and losing a dispute would cost it only fees, so it could refuse a valid withdrawal and hope Alice cannot play her side in time. Hence a reserve on each side, moved only by a lost dispute.

Each reserve deters one side and is the other side's prize. Write $S = V_{\max} + r_A + r_H$ for the contract, and ignore F, which only shrinks the prizes. Alice, with an honest balance β, probes with a claim of $V_{\max}$, which succeeds with chance p, the chance that the hub fails to answer in time: against a fold it gains her $V_{\max} - \beta + r_H$ and otherwise costs her $\beta + r_A$. The hub refuses a valid withdrawal of b, which succeeds with chance q, the chance that Alice fails to finish: it gains $b + r_A$ and otherwise costs r_H. In the worst case for each ($\beta = 0$; $b = V_{\max}$), neither pays when

$$p < \frac{r_A}{S} \qquad \text{and} \qquad q < \frac{r_H}{S}.$$

Both must hold at once, since the terms are fixed at opening, before anyone knows which side, if either, will cheat; and the reserves cannot be set independently, since raising r_H to deter a refusing hub also raises the prize for probing it. Together the two conditions require

$$(r_A + r_H)(1 - p - q) > (p + q)\, V_{\max}.$$

For $p + q < 1$ the reserves must total more than $(p + q)V_{\max}/(1 - p - q)$, about $(p + q)V_{\max}$ when both are small, at the least $r_A = pV_{\max}/(1 - p - q)$ and $r_H = qV_{\max}/(1 - p - q)$. That grows without bound as $p + q$ approaches 1, and for $p + q \geq 1$ no reserves deter both. Deterring $p = q = 10\%$ takes $V_{\max}/8$ on each side; $p = q = 1/3$ takes $V_{\max}$ on each side, twice the cap locked beside it.

Reserves that large are rarely worth locking. What sets a floor is fees: a lost dispute should cost the loser at least the other side's fees. Part of those is known when the contract is signed: the fee floors of its pre-signed transactions, 30,981 sats in the deepest dispute under BABE [4, §8], and 197,356 under BitVMX [3, §6]. So neither reserve should be smaller, and lichen's hub takes for its own the larger of its share of the cap (2% by default) and that figure. At $K = 24$ that deters refusals with less than about a 2% chance; Alice's reserve at BABE's floor deters probes with less than about 0.2%, at BitVMX's about 1%. In practice, then, the reserves pay for the dispute, and what deters is liveness: the hub's own, which keeps p small, and Alice's tower (Section 5.3), which keeps q small.

Two cautions on p. It is not the hub's average downtime but its chance of missing a whole response window at a moment Alice chooses: she can wait for an outage she can see, or cause one, though it must then outlast the window, which the hub can make hard with redundant signers of its own. And an outage is shared by all of the hub's contracts, so its users can all probe at once: what an outage puts at risk is the sum over its contracts of $V_{\max} - \beta + r_H$, not one contract's worth, part of the mass-exit problem of Section 8.

Who it suits. A reserve is locked, not spent: an honest party loses it only by losing a dispute, which for an honest Alice means going absent in the middle of one. The binding cost falls on the hub, whose reserve must cover Alice's worst-case dispute however small her balance. Under BABE that is at most about 31 kvB, 31,000 sats at 1 sat/vB and 310,000 at 10, so the contract suits balances from about a hundred thousand sats; under BitVMX, about 117 kvB, 117,000 sats at 1 sat/vB and 1.2 million at 10, so balances from about half a million. Large balances add no cost: the dispute does not grow with b. Amounts are 32-bit counts of one satoshi, so a contract holds up to about 42.9 BTC.

5.3. User liveness and watchtowers

The user needs to be online at two times only: before T_{close}, to withdraw or renew, and during a dispute, which only she can start. How demanding the dispute is depends on the gadget. Under BABE she signs b for her request, and, once the hub has signed it too, her proof's first element for her assertion, due only by a deadline shortly before T_{close}. Her one move under time pressure is her answer to a challenge: a hash preimage, posted in a transaction she signed when the contract opened. Under BitVMX it is a dozen moves over hours (days on mainnet), each due within a window, and each is her one-time signature on values that depend on the hub's last move, so they cannot be signed in advance and handed to a watchtower, as Lightning's penalty transactions are.

Because a standalone contract gives the hub nothing to broadcast against her, Alice can delegate in a way Lightning cannot: to a *tower* of her own, started only when a withdrawal is refused and discarded after. Under BABE it holds signatures only, hers on b, on her proof's first element and on the answer, and her fee budget's key: no key that signs a move, so at worst it stays silent, and several can run at once, run by strangers. One played a dispute on Mutinynet from lichen's Docker image on a rented machine, while Alice's own device did nothing from handing it the dispute to sweeping her payout [4, §7]. Under BitVMX it holds her contract's pre-signed transactions and the keys for her moves, but not the key to her amount, so it can play her side, and at worst play it badly, but cannot claim a different amount; lichen's BitVMX tower proved the withdrawal and played a hostile hub to the payout on Mutinynet in about 75 minutes [3, §10]. Embedding the contract in a Lightning channel gives this up: a channel's revoked states bring back the need for an always-on watchtower.

6. Finality from Bitcoin

With a single, or indeed a distributed, sequencer signing updates to L2 state, without a consensus mechanism like proof of work or proof of stake, it's free to "rewrite history" for transactions on the L2, enabling double spend attacks. The techniques used in ETH L2s to handle this involve the rollup data publication and contract state on the base chain; that is not available here. So in the absence of proposing expensive consensus on L2, it makes the most sense to inherit finality from Bitcoin, as Shielded CSV [24] and Shielded Bitcoin [25] do (the first by publishing each transaction's nullifier on Bitcoin, the second by publishing every transaction there in full), but here using the single use seal concept [20], with one commitment per batch.

If each sealed batch's root is committed in a Bitcoin transaction that spends the previous batch's commitment output, a fork of the layer 2 needs a double spend on Bitcoin. It costs one small transaction per batch, not per layer 2 transaction. A committee of sequencers then needs only

one honest member for inclusion (members append batches in turn); forks are excluded by Bitcoin, not by a threshold.

What sealing does not provide is availability: a seal commits to a root, not to the data behind it. This paper does not answer this question; it is a function of exactly how the L2 ledger is constructed (with a Bitcoin publication component, or not). But if data availability is not addressed, the user does have to trust the sequencer in a very concrete sense: if the sequencer refuses to provide the Merkle path for her return note, the user has no way to retrieve their funds in the contract, and loses them. This strongly argues for the above-mentioned "committee of sequencers", because you just need 1 to include your withdrawal and give you its path to be safe[4].

What the withdrawal statement checks. The sequencer's anchors form one chain of spends on Bitcoin: each spends the output the previous one created, starting from the layer 2's *genesis seal*, and commits to the history of state roots up to its height. Each anchor is recorded on L2, and the chain proof checks it with the rest: that its transaction spends the previous seal, and that it sits in the block it names, by a Merkle path to that block's header. When Alice withdraws (in the currently implemented design) she provides a root that has been anchored (not necessarily the most recent) along with a withdrawal transaction she performed, which is sufficient for proof, because the anchors provide a unique history (there is no other history, we claim, where she keeps the money on L2 while withdrawing).

The withdrawal statement also requires that Bitcoin headers follow the anchor's block, each linking to the one before and meeting its own difficulty target, with total work at least W. The genesis seal and W are part of the contract's identifier $\mathcal{D}$, so the contract fixes both: W is set when the contract opens, as ten blocks' work at the difficulty of that moment. A header declares its own difficulty, but that gains a forger nothing: work is summed, and producing headers worth W in total costs about W hashes however the difficulty is split among them. So no retarget rule needs checking, and the contract's length does not matter.

The cost is small. About ten headers, each two or three SHA-256 compressions, which RISC Zero accelerates: a few thousand cycles of a withdrawal statement that costs 389,020 cycles in all. For the sequencer, the chain proof's check of an anchor record is about 135,000 cycles. The dispute is unchanged. In the cooperative case, the hub checks the anchors with its own node.

Forgery then becomes a matter of mining a fork in Bitcoin itself. In current Bridge designs, like Union [16], a challenger answers the claimant's chain with a heavier one, which the claimant may dispute in turn, in a second BitVMX game with the roles reversed.

But we don't want to or need to do that; the claimant is always the user and the dispute is settled on one statement.

So any forgery attack is on the side of the user; and such an attack to achieve maximum returns *could* use it to steal from the hub across multiple contracts. So the bound is not one contract's cap but the hub's total headroom exposed to a single forged chain. As long as that stays well below the reward of the blocks a fork must replace, the attack remains un-economic. This is the

[4] A more advanced idea is publishing withdrawals on Bitcoin, requiring L2 to read L1 in a similar way to what is described next

kind of security you can claim for Bitcoin itself, basically.

7. Exiting against a hostile operator

A withdrawal starts with a return to the hub on the L2, which the sequencer must include. If the hub and the sequencer are one operator, it profits from not including it, since whatever isn't withdrawn by T_{close} is the hub's: it may censor the return, stop sealing, withhold data, or vanish. The exit doesn't start with a return.

The claim. Alice claims from a state S already anchored on Bitcoin: "at S these notes of mine were unspent; pay me b". She needs only her *exit kit*, which the sequencer serves after each of her spends and receipts and her wallet checks against the signed root: each note's path in the commitment tree at S, and its nullifier's absence from the spent set there. A kit never goes stale, since the tree only grows. Her claim's fingerprint $E = H(c \,\|\, O_D \,\|\, b \,\|\, h_S \,\|\, SR_S \,\|\, \textit{nullifiers})$ names the contract by its deposit coin O_D, which only its funding can spend; each note's owner signs E, and her `request` posts the claim with the signatures (about 1.2 kB at most), so that the hub can search for the notes and cite the signatures.

The counter. The hub may answer only with a *counter*, on one of two grounds, each something only Alice could have done:

- a note spent at an anchored state after S, in the one anchored history, whose anchor confirmed in or before her request's block. A later one could be a transaction of hers the operator held and sealed after she committed to her claim. Her own return to the contract, of at least b, never counts: the hub would hold her tokens, and the contract still pays once;
- her signed claim of one of the notes through another contract.

A nullifier enters the spent set only by a spend, which needs the note's keys, or by an *exit record*, which the hub posts after a claim to retire its notes and which the chain proof accepts only with the owners' signatures on that claim. So the hub can't forge a counter.

On BABE. The claim and the counter ride on the dispute's first two transactions. `request` signs E beside b; `respond` signs b and E in one of two leaves, plainly or with the counter K, a Groth16 proof of the counter guest, signed whole. Alice's statement, fixed at setup, has three branches: a return, as before; her claim, after a plain response; or K not being a counter, after a countering one. A true counter leaves her no branch, and the hub takes the contract; a false one pays her, and that branch's proof needs no secret of hers, so anyone can make it. Because `respond` precedes `assert`, a counter can't wait for an exit record of another claim: the second ground, her signature, is what stops two requests through two contracts in one block from both being paid. The companion paper gives the graph, the sizes and the cost [4]; the hub's window w_H grows to two hours on Mutinynet, so that it can prove a counter.

What it doesn't solve. The exit is safe exactly when Alice can read the anchored data she needs: an operator that seals a spend of hers and withholds its new notes' data leaves her old notes countered and the new ones without a kit. Requiring the sender's acknowledgment of the data before a spend can counter her is the planned remedy. Shielded notes can't exit yet, since

their owner's authority is knowledge of a key inside the proof, with nothing to post; nor can a tower play an exit without her keys, since her proof depends on the hub's response.

8. Scaling with multiple hubs

The worst failure mode for a Lichen-style L2 is mass exit. As we know even in Lightning there is the idea that, if the number of users is high enough, mass publication of old/invalid states could lead to an excessive reliance on prompt confirmation on a blockchain where space is a scarce resource.

Now imagine lichen similarly, but with that problem exacerbated: a hub holding 60,000 users (note: the problem is not *necessarily* any better with the 60k spread over multiple hubs). Now each withdrawal may take a number of transactions and a chunk of vB, with the numbers depending on the verifier gadget used. It's clearly the case that a malicious hub can trigger a very large block space demand and, with a lot of difficulty to be fair, *could* steal from many honest users at once depending on the order in which transactions get confirmed. The same attack exists in reverse: if a user-attack generates a ton of Sybil contracts with one hub, they could trigger mass exit with fraudulent exit requests.

It is worth mentioning that the more likely failure mode — that is of hub infrastructure failure leading to offlineness — tends to use up less onchain space with any verifier gadget used for the onchain transactions, but it can still be quite significant, even here.

Two things must be chosen to get a sense of the potential safe scale of a Lichen L2: how long is the dispute phase allowed to last? And, how much fees must be held in reserve by both parties to ensure that the dispute can fully play out in time? Given this data, some specific, though estimated, numbers of how many users a hub can support, can be generated. These numbers are laid out in [3] and [4].

We can improve these figures on *some* axes by having multiple hubs. If we had 20 distinct hubs of 3000 users and not 1 hub of 60,000 it would reduce some risk for users in particular. Which leads to the question:

Can you pay users in a different hub? Certainly you can, with some infrastructure. There's more than one way to do it, but for brevity we describe the simplest: if Hub 1 and Hub 2 exist, with their own distinct ledgers, Alice on Hub 1 can pay Bob on Hub 2, in bitcoin, if Hub 2 has a contract on Hub 1 with inbound capacity. She simply pays Hub2's address on Hub 1's ledger, contingent on an atomic swap set up with the payment to Bob on Hub 2's ledger. There could also be routing, but the principle is clear.

Different ledgers? It seems natural in this multi-hub world to allow users to use a Hub whose ledger rules (e.g. does it use shielded transactions or smart contracts or ..) can be distinct and whose consensus establishing mechanisms are distinct. Alternatively they could all rely on the same finality and proof of validity, in one ledger, as distinct assets.

9. Costs

Many costs will be specific to the verifier gadget chosen, so we point the reader at [3] and [4]. However some costs come from the generic Risc0 architecture chosen:

Can a user dispute on her own hardware? A dispute needs a non-urgent (only T_{close} bounds a withdrawal) but non-trivial computation from Alice (though nothing is needed in the cooperative case, where the hub reads the layer 2 directly). That is, a proof of her withdrawal. As built, the statement costs 389,020 zkVM cycles, one segment, using RISC Zero's accelerators (about 10 cycles for a 256-bit modular multiplication, 68 for a SHA-256 block [21]); without them it was 5.9 million, almost all of them in the sequencer's signature. Proving it takes 74 s on an Apple M4 Pro and 359 s on a 4-vCPU x86 server, which smaller segments reduce [22]; a CUDA GPU would be faster still. BABE's statement, which also checks the hub's signature on b, runs in about 480,000 cycles, still one segment, and proves in 640–660 s on that server; under BABE Alice proves only after the hub has signed b on chain, and has until shortly before T_{close} to do it. The Groth16 wrap is a fixed cost of a few minutes with 8 GB of memory at peak, today on x86 only; it can be delegated, since the wrapper sees only the receipt and its public journal. A phone is plausible for the proof with small segments and a delegated wrap.

Opening a contract Under BABE, opening costs the hub about ten seconds of garbling and Alice a download of about 240 MiB, the instances she keeps until the contract closes; both are paid again at renewal [4, §8].

Onchain costs For the anchoring of L2 state updates, there is an anchor per batch, 269 vB, including the OP_RETURN output for SR. For each individual contract, a fold costs 193 vB. Even for faster L2s with more aggressive finality, a cost of 269 vB every block or few blocks is trivial. A dispute under BABE is 9.7 kvB if the hub does not challenge, and at most about 31 kvB if it does (Table 1).

10. Positioning

We have already briefly discussed how this is different from just "payments over Lightning via a hub" in the Introduction. We'll list here some key differences to some other existing L2 or L2-like Bitcoin constructions.

Against BitVM2 and Union. We consider BitVM2 [15], and Union [16], a bridge to Rootstock built on BitVMX by BitVMX's own authors. Both release bitcoin on Bitcoin against a claim about a program's execution, settled by fraud proof, with no change to consensus, and, like lichen, inherit the assumptions of a side system they do not secure. They differ in who the parties are, and most of the rest follows (Table 2).

The bridges pool their users' bitcoin, and a pool shared by users not yet known must be released by transactions that someone signs in advance on their behalf: BitVM2's committee, Union's functionaries. What then shapes the dispute is who may challenge a false claim. BitVM2 opens challenge to anyone, since anyone might be the one robbed; with the challengers unknown when the graph is signed, no interactive game can be pre-signed with them, so its dispute is a single round, in which the operator puts the verifier's whole intermediate state on chain at

	a BitVM2 bridge	Union	lichen
custody	one pool for all users	one pool, in fixed-size outputs, one per peg-in	a contract for each user, with the hub
pre-signed by	a committee, for every operator's graph on every deposit	the N functionaries, a two-party game for every pair of them on every peg-in	the user and the hub
setup trust	one honest committee member	one honest functionary	none
the claimant	an operator, who paid the user from its own funds and claims reimbursement	a functionary, likewise	the user, for her own withdrawal
the challengers	anyone; safety needs one honest one watching	the other functionaries; safety needs one honest one watching	the hub alone, the only party a false claim can rob
the dispute	one round: the operator asserts the verifier's intermediate values, a challenger reveals one failing chunk (scripts up to 4 MB); 42 blocks	BitVMX's interactive search, 4-ary: about 270 kvB at worst; a heavier chain is answered in a nested game	under BABE, at most five pre-signed transactions, about 30 kvB, whatever the claim
the user's part in a dispute	none: she was paid up front	none, likewise	a party: she plays her side, or her tower does, holding none of her keys
a refused withdrawal	it waits for a willing operator	it waits for a willing functionary	she enforces it on chain herself
stakes	operators' collateral; challengers fund the assert	bonds sized to dispute costs, shared by a packet of peg-ins	the contract reserves r_A, r_H
liquidity	operators front withdrawals until reimbursed	functionaries likewise, one peg-out each at a time	the hub locks $V_{\max} - D + r_H$ for each user until T_{close}
the asset	bridged bitcoin, fixed amounts	bridged bitcoin, fixed amounts	variable amount bitcoin within each user's cap, credit beyond it
expiry	none	none	T_{close}: withdraw, or renew, before it

Table 2: Two bridges and lichen. The bridges' figures are their authors' estimates; lichen's depend on the gadget; see [3], [4] for measurements and estimates.

once. Union closes challenge to its functionaries, who are known at setup, and so can pre-sign BitVMX's interactive search between every pair of them, which puts less data on chain over many more rounds. Its price is that safety needs an honest functionary not only at setup but online whenever a claim is made, and that the pairwise games, and the bonds that fund them, grow with the number of functionaries. A lichen contract is not pooled: it has two parties, both present when it opens. They pre-sign it themselves, so there is no committee and no setup trust; only the hub can be robbed by a false claim, so only the hub challenges; and the dispute between them needs no search: a challenge, answered from garbled circuits Alice checked when the contract opened (under BitVMX, it is Union's game with a single pair). Splitting the pool has its own price: the hub's locked liquidity, a token that is bitcoin only within each user's cap, an expiry, and a user who must take part in her own dispute. **The bridges ask their users to trust one signer at setup and one challenger afterwards, and to be present for none of it; lichen asks them to trust no one, but to be present for their own disputes**.

Other designs, briefly.

- *Ecash* (Cashu, Fedimint): lichen removes the mint's power to take the funds, within each user's cap, at the price of the hub's locked liquidity; the mint may print as it likes, but every redemption is against its own collateral.
- *Ark*: a shared tree of outputs with an operator and an expiry, where a lichen contract is per user; both ask users to act before an expiry. An Ark balance is a claim on a specific pre-signed output, exitable by its holder alone; a lichen balance is held in sats on a programmable ledger and redeemed one-for-one in bitcoin from the hub's collateral, without the hub's cooperation, up to the holder's contract cap, and is credit on the hub beyond it.
- *Payment pools* (one shared output, balances updated by all members): need every member's signature for an update and a covenant (or an exponential tree) for unilateral exit; a lichen transfer needs only its sender, and exit needs no consensus change.

11. Implementation

The design is implemented in proof-of-concept form as two code bases, LN-GAP [8] and Lichen [9], with tests and demos executed on bitcoin regtest and on mutinynet. Any bitcoin-related data in this paper come from the concrete executions there, including playing out disputes fully.

The implementation uses the BABE gadget [4] by default, with the BitVMX gadget [3] also built, and a shielded-notes L2 for private offchain transfers. Validity proofs are implemented, along with anchors in bitcoin transactions for finality. Under BABE, three endings of a dispute have been played on Mutinynet with a real RISC Zero proof (a true claim challenged and answered, a true claim unchallenged and played by a tower, and a false claim), and the others on regtest.

The exit (Section 7), for a user whose sequencer censors or vanishes, is built under BABE and played on regtest on the mock statement: an honest exit paid, a stale claim countered, one note claimed through two contracts and the second countered, a forged counter proved false. It is not yet played with real proofs, for shielded notes or by a tower, and the sender's acknowledgment that would close its data-availability gap is not built. Under BABE, Alice's download at each opening, about 240 MiB, is the largest cost to reduce; and the anchors through which a tower

would bump her transactions' fees need a key of their own.

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